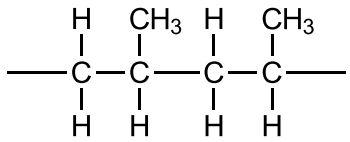
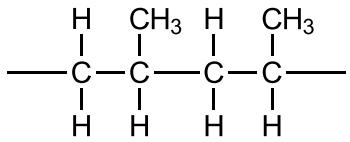
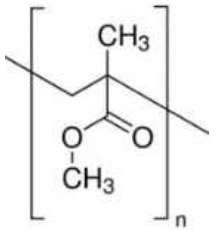
4(a)(i)	reduction	**1**
4(a)(iv)	conc. H_2SO_4 **OR** conc. H_3PO_4 **OR** Al_2O_3 **AND** heat	**1**
4(b)(iii)	 **M1** **M2** optical isomerism **M3** the methylated carbon has four different groups attached	**3**
4(b)(iv)	**M1** non-biodegradability **M2** harmful combustion products	**2**

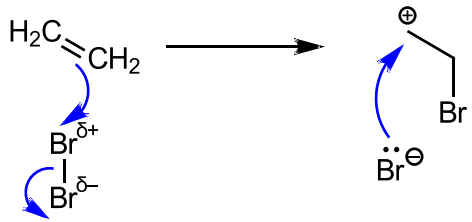
Examiner report missing
9701_w25_qp_23 Question 4

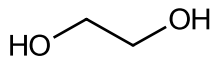
4(a)(i)	reduction	**1**
4(a)(iv)	conc. H_2SO_4 **OR** conc. H_3PO_4 **OR** Al_2O_3 **AND** heat	**1**
4(b)(iii)	 **M1** **M2** optical isomerism **M3** the methylated carbon has four different groups attached	**3**
4(b)(iv)	**M1** non-biodegradability **M2** harmful combustion products	**2**

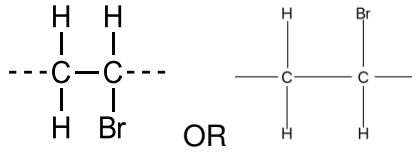
Examiner report missing
9701_w25_qp_21 Question 4

5(a)(i)	C_2H_3O	**1**
5(b)(v)	• **A** C–H alkane • **B** C=O ester • **C** C=C alkene • **D** C–O ester 2 points = 1 mark, all 4 points = 2 marks	**2**
5(b)(vi)		**1**

Examiner report missing
9701_s25_qp_23 Question 5

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**

4(c)(i)		**1**
	bromoethene	**1**

4(c)(ii)	 OR	**1**

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

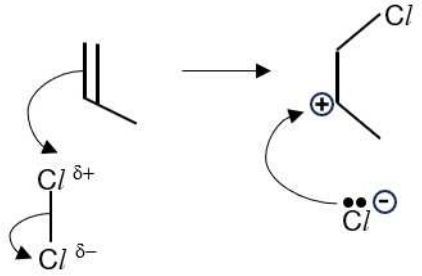
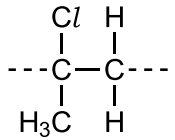
(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

2(a)	<table><tr><td>formula of chloride</td><td>● $NaCl$</td><td>● $MgCl_2$</td><td>● $SiCl_4$</td></tr><tr><td>structure</td><td>giant</td><td>● giant</td><td>● simple</td></tr><tr><td>bonding of chloride</td><td>● ionic</td><td>● ionic</td><td>covalent</td></tr><tr><td>pH of solution formed</td><td>● 7</td><td>6.2</td><td>● 1–4</td></tr></table>	**3**
2(d)(i)	 ● curly arrow from C=C bond to $Cl^{(\delta+)}$	**1**
	● correct dipole on Cl_2 **AND** curly arrow from Cl–Cl bond to $Cl^{\delta-}$	**1**
	● correct carbocation intermediate	**1**
	● curly arrow from lone pair on Cl^- to C^+	**1**
2(d)(iii)		**1**

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

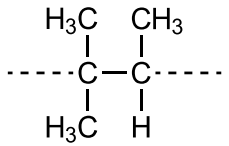
(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

4(a(i)	3-methylbut-1-ene	**1**

4(a)(v)		**1**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

Many candidates appreciated that methyl/alkyl groups are responsible for a positive inductive effect and that the more alkyl groups, surrounding the carbocation, the more stable is the carbocation intermediate. However, the application of this concept to the intermediates of C and D was not well recognised.

Common errors were statements that the tertiary halogenoalkane was more stable, in C, and the secondary halogenoalkane was more stable, in D, rather than their respective intermediate carbocations.

Many candidates quoted Markovnikov's rule without any further explanation.

(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

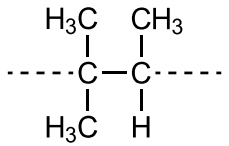
The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified <u>concentrated</u> $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed <u>concentrated</u> H_2SO_4 (as a catalyst).

4(a(i)	3-methylbut-1-ene	**1**

4(a)(v)		**1**

Question 4

(a) (i) Most candidates correctly gave the systematic name of the alkene **D**, with a molecular formula of C_5H_{10}, as **3-methylbut-1-ene**.

(ii) Few candidates gave a correct explanation of why the alkenes, **C** and **D**, did not exhibit geometric isomerism.

In **C**, one C (in the C=C bond) has two methyl groups and in **D**, one C (in the C=C) has two hydrogen atoms; neither will exhibit geometrical isomerism.

(iii) Many candidates gave pent-2-ene, or **C** or **D** as examples of a positional isomer.

(iv) When **D** is reacted with $H_2(g)$, 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$, is formed. Many candidates omitted the brackets surrounding the side-chain methyl group, on the CH group.

(v) Many candidates gave a monomer containing only two methyl groups or structures that still contained a double bond.

(b) The question contained information regarding the relative intensities of the M and (M + 1) ions, from a mass spectrum of **C**, which showed a molecular ion peak at 70.

Many candidates correctly calculated that C contained 5C atoms but then failed to use this information, in conjunction with the molecular ion peak, 70, to confirm that the molecular formula of C was C_5H_{10}.

(c) (i) The mechanism for the addition of HBr to **C,** 2-methylbut-2-ene, was quite well answered. Common errors included full charges on the H-Br, instead of a dipole, curly arrows, from C=C to $H^{\delta+}$ of HBr, arrows starting at one end of the C=C bond and intermediates with both a –ve charge and a partial negative charge.

(ii) This question was well answered with a common error being the use of C_3H_7, instead of the unambiguous expanded version, $(CH_3)_2CH–$.

(iii) This question required an explanation as to why different major products were formed when HBr reacts separately with C and D.

Many candidates appreciated that methyl/alkyl groups are responsible for a positive inductive effect and that the more alkyl groups, surrounding the carbocation, the more stable is the carbocation intermediate. However, the application of this concept to the intermediates of C and D was not well recognised.

Common errors were statements that the tertiary halogenoalkane was more stable, in C, and the secondary halogenoalkane was more stable, in D, rather than their respective intermediate carbocations.

Many candidates quoted Markovnikov's rule without any further explanation.

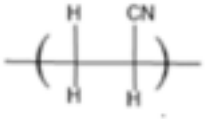
(d) *Please note that due to an issue with diagram labelling in this question, full marks have been awarded to all candidates to make sure that no candidates were disadvantaged. This has been corrected in the published version of the paper.*

The question required completion of a table of reagents and conditions for a reaction scheme.

The formation of alcohol, **F**, from a halogenoalkane, **E**, required the use of NaOH(aq).

Cleavage of the double bond, in **C**, to form a carboxylic acid, **G**, and propanone, was carried out using hot acidified concentrated $KMnO_4$.

The formation of ester, **H**, from **F** and **G**, needed concentrated H_2SO_4 (as a catalyst).

6(a)	 OR 	**1**

Question 6

(a) The single repeat unit derived from the monomer, CH_2=CHCN, by addition polymerisation was quite well answered.

(b) (i) Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

(ii) Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the CN^- to the C (in the C=O bond) did not start from the lone pair on the C of the CN^- ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the CN^- nucleophile and many answers incorrectly showed an arrow from an H^+ ion to an O^- (in the intermediate).

(iii) Some good answers were seen.

(iv) The absorptions in the infrared spectrum of CH_2=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

(c) (i) Very few candidates gave an accurate description.

(ii) A number of answers were awarded some credit. Few candidates were able to give a full description.

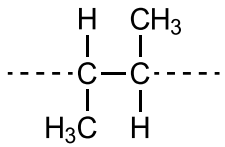
(d) Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

(e) The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

(f) (i) It was well known that the name of the type of reaction in step 1 is substitution.

(ii) Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	**1**

4(a)(ii)		**1**

Question 4

This was a second organic chemistry question, which included some analytical work. This was generally answered well by many candidates, though later parts on IR and MS appeared less familiar to some.

(a) (i) 'Heat' or equivalent was often missing. H_2SO_4 was a popular answer despite the conditions suggested in the syllabus.

(ii) Many repeat units were shown containing four carbon atoms, as if the CH_3 groups had been incorporated into the backbone. Some candidates showed an ethene structure with missing CH_3 side groups.

(iii) Candidates very often showed the reverse equation with C_4H_8 and H_2O. Quite a few answers contained Al_2O_3 as a reactant and several answers omitted H_2O as a product.

(b) (i) This was well answered.

(ii) There were many correct structures/names for iodoform but very few correct propanoate ions.

(c) (i) There were many correct answers to this item.

(ii) Catalyst and oxidising agent were the common errors here, but otherwise there were many correct answers.

(d) (i) Compound **F** was commonly given incorrectly as the unknown compound, referring to an O–H absorption in the range 2850–3000 cm^{-1}. Candidates are reminded that C–H stretches appear in all organic spectra. Where absorptions are referenced, candidates need to identify which bond is stretching: 'nitrile', for example, is insufficient.

(ii) There were many null responses to this. Where attempted, candidates' answers were evenly split between correct identification of an identical abundance and some complicated (erroneous) mathematics.